NP Employee Assistant

AI Engineer Assessment — Design Report

A local-first retrieval-augmented assistant over 10,018 employee records, with FastAPI, source citations, reproducible preprocessing, and optional OpenAI-compatible or Ollama generation.

1. Architecture

employee_np.xlsx	→	Clean + validate	→	JSONL documents
User question	→	FastAPI /ask	→	TF-IDF vector search
Top-k sources	→	Grounded prompt	→	Generator + citations

Request lifecycle

At startup, the workbook is validated and converted into one self-contained text document per employee. TF-IDF creates sparse embeddings stored in a local Joblib index. POST /ask validates a question, retrieves top-k records by cosine similarity, applies a minimum relevance threshold, and passes only those records to the configured answer generator. The response exposes its sources.

2. Data ingestion and processing

Required columns are validated; duplicate employee IDs are removed; whitespace and missing text are normalized; numeric fields are coerced; and Excel serial or datetime hire dates become ISO dates. Each row remains one chunk because it is already a coherent factual unit. Processed records are stored as JSONL.

3. AI/ML implementation

TF-IDF word and bigram embeddings form a transparent offline baseline that performs well for names, IDs, roles, and departments. Cosine similarity ranks relevant records. The default extractive mode works without secrets. OpenAI-compatible and Ollama adapters enable generative RAG; temperature is zero and prompts prohibit unsupported claims.

Evaluation

The evaluation script derives labeled department and job-title queries and reports Precision@5 and Mean Reciprocal Rank. Unit and API tests cover document creation, retrieval relevance, health checks, valid questions, and input validation.

4. API layer

FastAPI provides GET /health and POST /ask, automatic OpenAPI/Swagger documentation, typed request and response schemas, bounded top-k input, whitespace normalization, validation errors, and safe handling of upstream model failures.

5. Limitations

TF-IDF is lexical and may miss paraphrases. Top-k retrieval cannot safely calculate exact whole-dataset aggregates. Indexing currently rebuilds at startup. LLM prompt constraints cannot eliminate hallucinations. Authentication, authorization, audit logging, and field-level salary permissions are outside this prototype.

Accuracy and hallucination improvements

Use dense domain-tested embeddings, hybrid search, metadata filtering, and a reranker. Route count and average questions to validated structured queries. Require citations, reject low-confidence answers, build a human-labeled evaluation set, and evaluate faithfulness after every model or prompt change.

Production deployment

Deploy the Docker image through CI behind an authenticated API gateway. Move data to PostgreSQL and vectors to pgvector or a managed vector database. Schedule ingestion separately, version indexes, store secrets in a secret manager, use TLS and rate limits, and enforce role-based access to sensitive employee fields.

Monitoring

Track latency, errors, request volume, provider cost, empty-result rate, retrieval scores, source use, and feedback. Monitor freshness, missing values, duplicates, schema and distribution drift. Run regression evaluations for every data, model, embedding, or prompt release and alert on deterioration.